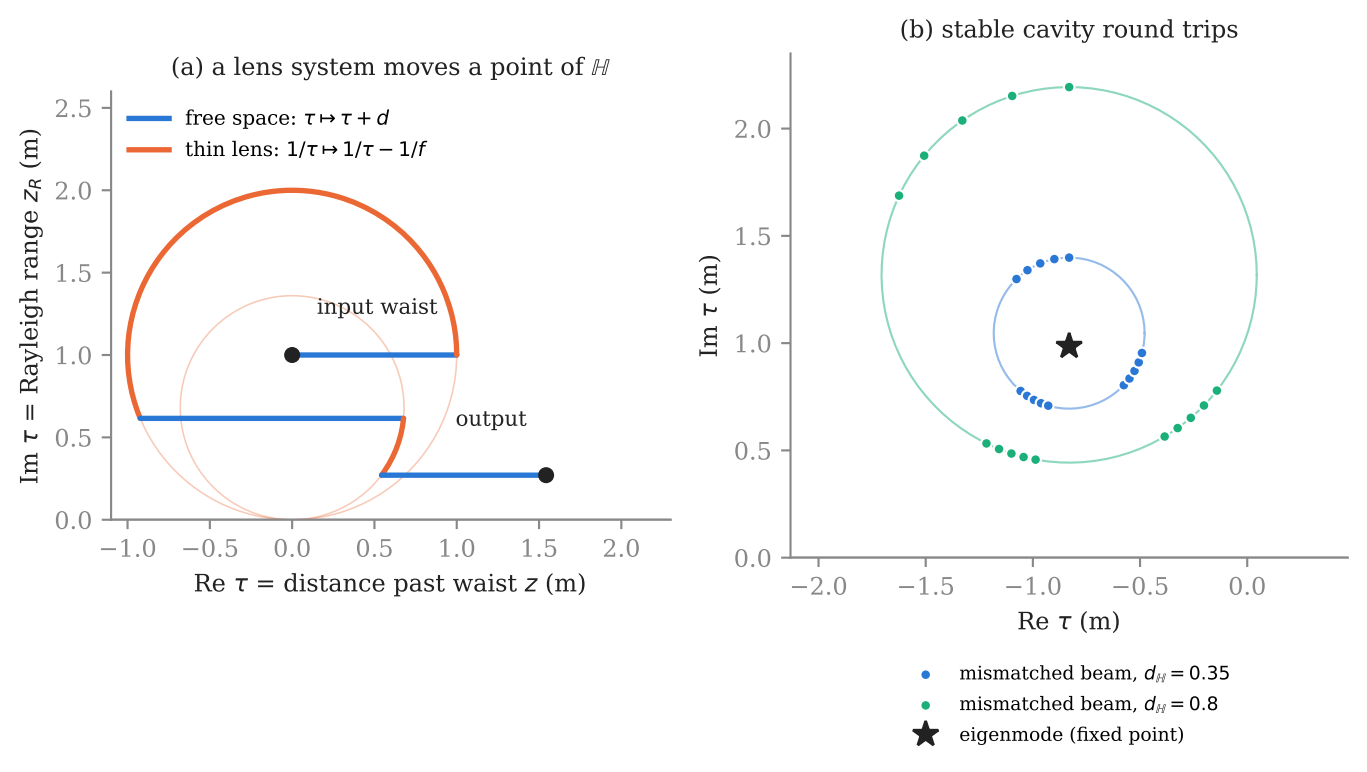
(a) a lens system moves a point of $\mathbb{H}$
free space: $\tau \mapsto \tau + d$
thin lens: $1/\tau \mapsto 1/\tau - 1/f$
input waist
output
Im τ = Rayleigh range z_R (m)
Re τ = distance past waist z (m)
(b) stable cavity round trips
Im τ (m)
Re τ (m)
mismatched beam, $d_{\mathbb{H}} = 0.35$
mismatched beam, $d_{\mathbb{H}} = 0.8$
eigenmode (fixed point)